Show Me: Prestige and Pool Effects

Advisor-ready summary deck (8–12 slides)

---

1. Research question and framing

- Do **prestige assignments** and **senior-rater pool context** change promotion outcomes?
- "Big fish / strong pond" framing for both **prestige** and **pool** effects
- Population: YGs 2002–2011 (with targeted branch filters)

---

2. Methods (one slide)

- Cox proportional hazards with time-varying covariates
- Key metrics are **z-scored** for comparability
- Interpretation uses **hazard ratios** (relative promotion risk)

---

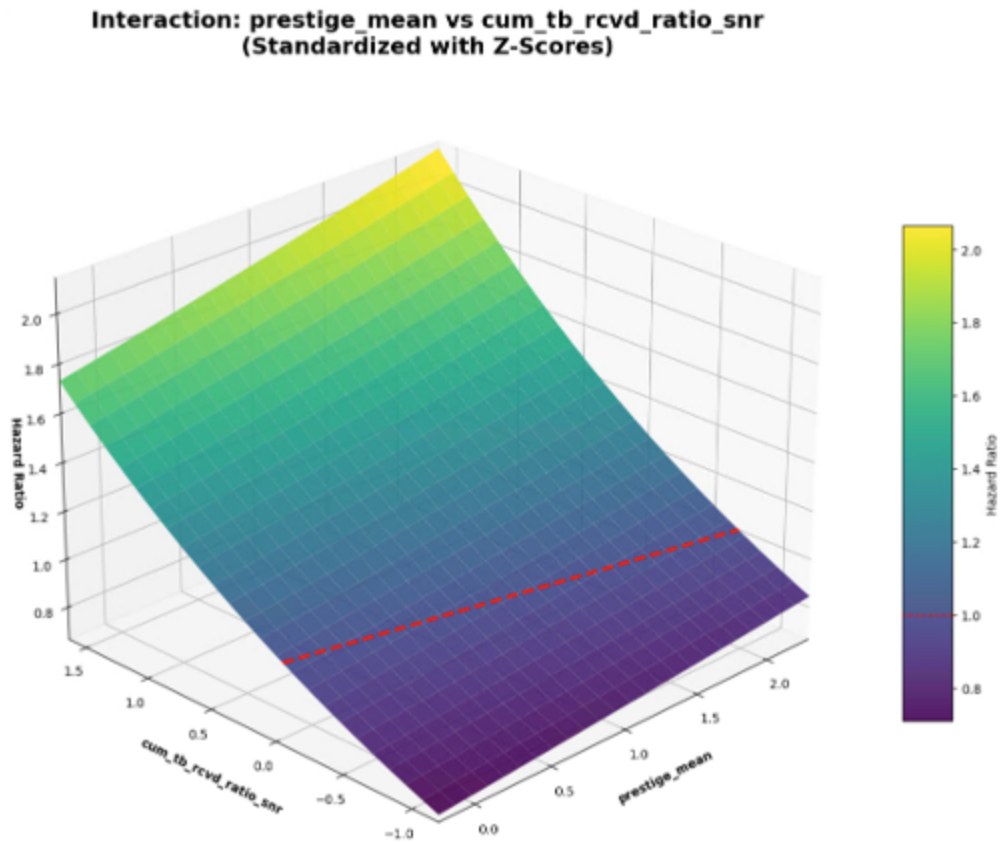
3. Part 1 — Prestige story overview

- Focus: prestige exposure + cumulative TB ratio
- Runs used:
  - **Run 8**: no SF/Aviation (regular officers)
  - **Run 9**: IN/AR/FA only (combat arms)
  - **Run 4**: YGs 2002–2006 (cohort robustness)

---

4. Run 8 (regular officers): key findings

- `z_cum_tb_rcvd_ratio_snr` dominates promotion signal
- `z_prestige_mean` positive but modest
- **Negative interaction** → diminishing returns at high prestige + high TB
- Coefficients: 0.3235, 0.0783, **−0.0159** (interaction)

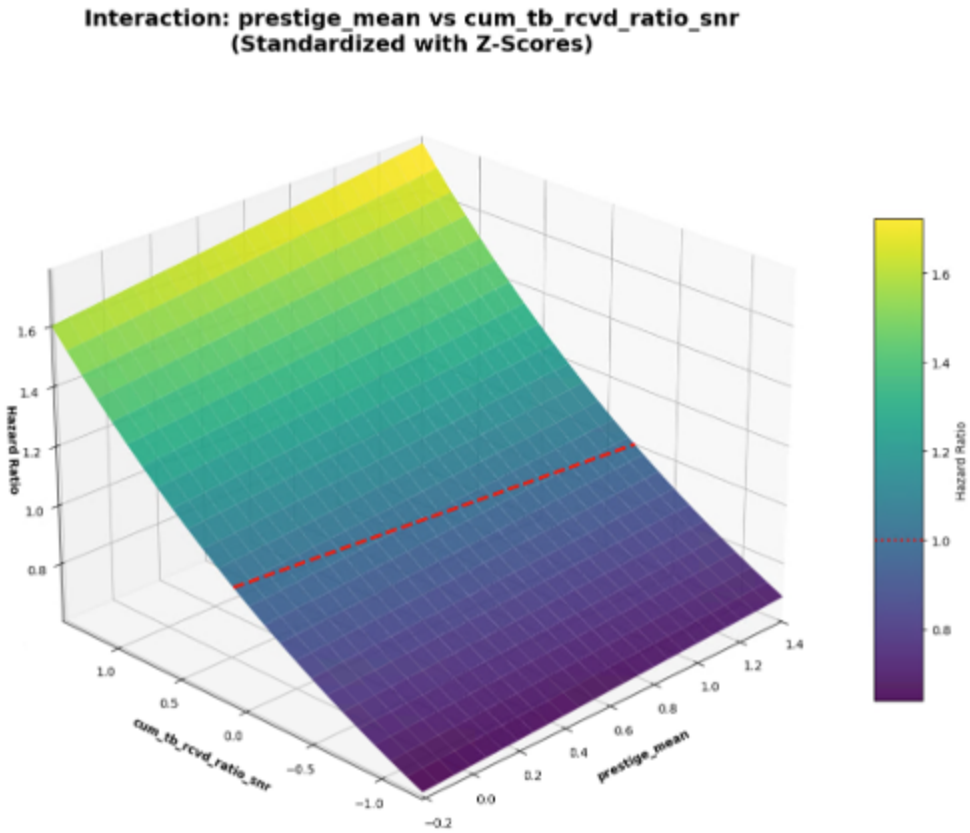


Run 8 interaction surface

---

5. Run 9 (combat arms): contrast

- TB ratio even more dominant
- Prestige effect smaller
- **Interaction flips slightly positive** → near-additive effects
- Coefficients: 0.3437, 0.0588, **+0.00815** (interaction)

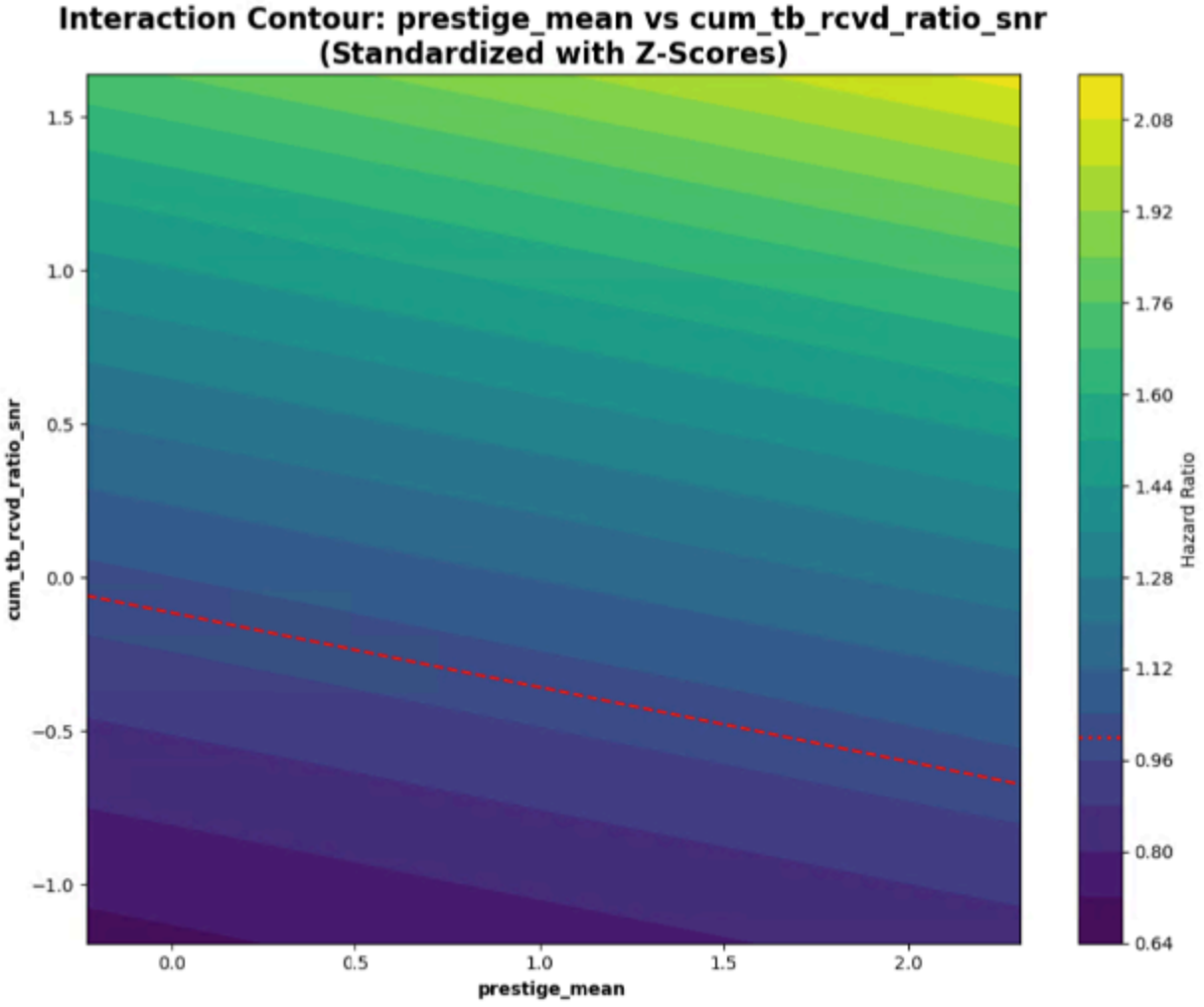


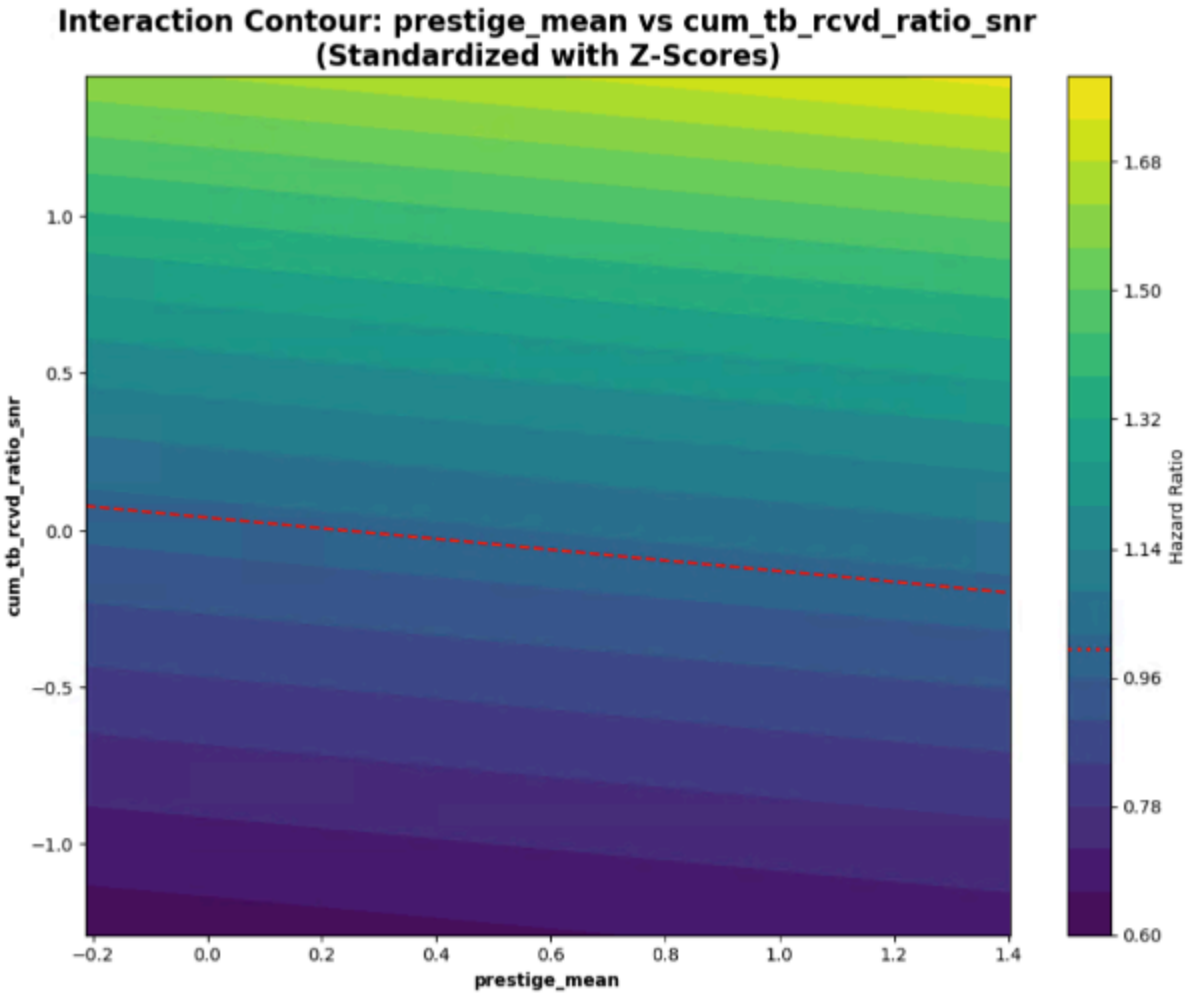
Run 9 interaction surface

---

6. Run 8 vs Run 9 (prestige takeaway)

- Both: TB ratio is the strongest driver
- Run 8: **dampening interaction** (strong-pond effect)
- Run 9: **near-additive** (minimal dampening)
- Interpretation: prestige effect depends on context and branch mix


Interaction Contour: prestige_mean vs cum_tb_rcvd_ratio_snr
(Standardized with Z-Scores)
cum_tb_rcvd_ratio_snr
prestige_mean
Hazard Ratio
1.5
1.0
0.5
0.0
−0.5
−1.0
0.0
0.5
1.0
1.5
2.0
2.08
1.92
1.76
1.60
1.44
1.28
1.12
0.96
0.80
0.64



---

7. Run 4 (cohort robustness)

- YGs 2002–2006 only
- **Stronger negative interaction** than Run 8
- Suggests dampening effect persists and may be stronger in earlier cohorts
- Coefficients: 0.2022, 0.0867, **−0.0293**

---

8. Part 2 — Pool story overview

- Two runs needed to handle collinearity in individual TB metrics
- Run 1: **absolute TB vs pool TB**
- Run 2: **rank-normalized TB vs pool TB**

---

9. Run 1 (absolute TB vs pool TB)

- Both main effects positive
- **Positive interaction** → strong pools amplify individual TB
- Pool TB mean is the larger main effect
- Coefficients: 0.2455, 0.3596, **+0.1079**

---

10. Run 2 (rank-normalized TB vs pool TB)

- Both main effects positive
- Interaction ~0 → near-additive
- Pool TB mean still dominant
- Coefficients: 0.3620, 0.5123, **+0.00335**

---

11. Pool story takeaway

- Pool strength matters across definitions of performance
- Interaction depends on how “individual TB” is measured
- Strong-pool amplification appears when using **absolute TB**

---

12. Bottom line (advisor summary)

- **Prestige effects are real but modest**
- **TB ratio is the strongest driver**
- Interaction sign flips by cohort/branch context
- Pool strength consistently matters; amplification appears in some definitions